7 Intravenous Cannulation (PIVC)

HME922 Procedural Station · 10 minutes · MM3 guideline + IVC DOPS form

guideline wording — say it aloud DOPS On the examiner's DOPS marking form ☐ tick when done

KEY MESSAGE

Only if needed. **Smallest gauge** that works (largest in trauma). **2% chlorhexidine / 70% alcohol — dry 30 s — no touch.** Max 2 attempts. Date the dressing. VIP every shift.

UNIVERSAL OPENING

- ☐ Hand hygiene — Moment 1 DOPS
- ☐ Introduce yourself: name + "medical student"
- ☐ Confirm ID — **3 identifiers: name, DOB, UR/address**; patient states, check ID band DOPS
- ☐ Explain procedure, purpose, risks, benefits DOPS
- ☐ Ask about **allergies** (tape, chlorhexidine, alcohol) DOPS
- ☐ Ask: any difficult cannulation before? ("took the anaesthetist 4 attempts")
- ☐ Allow questions DOPS
- ☐ Gain **informed verbal consent** DOPS
- ☐ Privacy, lighting, exposure; **raise bed height** DOPS

MEMORY AID — 5 MOMENTS

5 Moments of Hand Hygiene: 1 before touching patient · 2 before procedure · 3 after body-fluid risk · 4 after touching patient · 5 after touching surroundings

CLINICAL KNOWLEDGE — SAY IT

- ☐ > **3,500** cannula-related sepsis cases/year in Australia (~1.5 / 1,000 admissions) — devices bypass body defences
- ☐ Indications: medications · fluids/electrolytes · blood products · unstable patient access · IV contrast
- ☐ Delivery: continuous (N/S, Hartmann's) · intermittent (antibiotics, electrolytes) · bolus (ondansetron, metoclopramide)

AVOID if possible	NEVER (unless life-threatening)
Cubital fossa in a stable patient	AV fistula
Legs / feet in adults (DVT, phlebitis)	Burns, scars, injured areas
Previous IV sites, bruising	Infiltrated or phlebitic veins
Dominant arm	Infected areas
Joints — wrist, elbow	Dermatological conditions
Mastectomy / axillary clearance side	Sclerosed or thrombosed veins

CHOOSE THE GAUGE

Gauge	Colour	mL/min	Use
14 G	Orange	240	Trauma, massive resuscitation, major surgery, peri-partum, GI bleed
16 G	Grey	180	As above
18 G	Green	90	Blood, irritant drugs, rapid infusion, CT contrast
20 G	Pink	60	General: fluids, antibiotics, analgesia
22 G	Blue	36	Paediatrics, fragile veins
24 G	Yellow	20	Paediatrics, neonates

MEMORY AID — GAUGE COLOURS — "Old Grey Goats Pick Blue Yams"

Orange 14 · **G**rey 16 · **G**reen 18 · **P**ink 20 · **B**lue 22 · **Y**ellow 24
Bigger number = smaller cannula = slower flow.



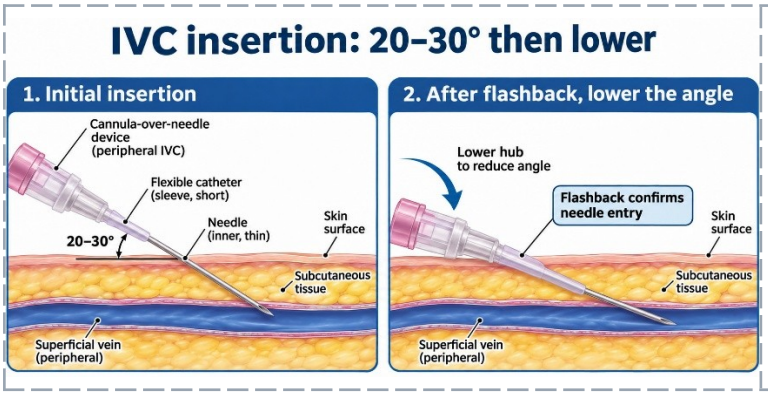
PREPARE

- ☐ Alcohol skin prep · IVC/dressing pack · tourniquet · transparent occlusive dressing · cannula · bung/valve · tape · bluey · gauze · sharps bin · splint/crepe if needed · 5–10 mL syringe · **0.9% saline flush** · blunt drawing-up needle · light · stool
- ☐ Standard precautions: hand hygiene, **gloves + safety glasses** DOPS
- ☐ Bluey under arm; you and patient both comfortable
- ☐ Vein: examine both arms; warm if needed; round, firm, elastic, well filled; straight length ≥ cannula

INSERT

MEMORY AID — KEY STEPS — "Clean, Dry, Bevel up, Flash, Lower, Slide, Flush"
Hand hygiene **before** opening sterile stock · ANTT throughout

1. Tourniquet on; locate vein DOPS
2. Skin prep **2% chlorhexidine in 70% alcohol** (aqueous if alcohol allergy); circular, inside → out ~5 cm; **dry 30 s**; clip hair, **never shave**; don't re-touch DOPS
3. Inspect cannula; rotate on stylet 360°
4. Anchor vein, skin taut, hand clear of insertion angle
5. **Bevel up**, with blood flow, **20–30°**, single smooth movement DOPS
6. **Flashback** → stop → lower angle almost to skin → advance a few mm DOPS
7. Slide **plastic cannula** off the needle to the hub — don't advance the needle; **never against resistance** DOPS
8. Release tourniquet; press over vein at cannula tip DOPS
9. Remove introducer → **straight into sharps**; attach bung while occluding DOPS
10. Bloods? Vacutainer onto bung, collect, then release tourniquet
11. **Flush 5–10 mL normal saline** DOPS
12. Transparent sterile dressing — **date/time sticker on it**; extra tape/splint DOPS
13. Failed? Retry only if confident + suitable vein — **max 2 attempts**



📷 Paste image here
Secured cannula with dated dressing

AFTER CARE — VIP SCORE

VIP	Findings	Action
0	Healthy	Observe
1	Slight pain OR slight redness	Observe
2	Two of: pain, erythema, swelling	**Resite**
3	All of: pain along vein, erythema, induration	Resite, consider treatment
4	Extensive + palpable venous cord	Resite, consider treatment
5	Extensive + **pyrexia**	Treat + resite (thrombophlebitis)

MEMORY AID — VIP
"**0–1 watch, 2 = resite**, 3–4 resite + think treatment, 5 = treat". Check **every shift**; remove when **no longer needed**.

☐ Complications: thrombophlebitis · thrombosis · extravasation (iron staining) · haematoma · nerve/tendon/ligament injury · infection — local (cellulitis) or systemic (CRBSI, endocarditis, osteomyelitis, sepsis, death)

UNIVERSAL CLOSING

☐ Introducer straight into sharps — **never recap** DOPS
☐ Waste: sharps / biohazard / general DOPS
☐ Area clean (no blood); patient comfortable; **bed back to lowest** DOPS
☐ Tell patient to report **pain, warmth, redness, swelling, bleeding** DOPS
☐ Hand hygiene — Moments 4/5 DOPS
☐ Document date, time, site, gauge, indication, clinician name DOPS
☐ Can't answer? "I'll pass that on to someone senior"
☐ Examiner asks **"How do you think that went?"** → one thing done well · one to improve · one strategy

MY NOTES